

Machines don't run on numbers. They run on parts, and the parts draw a shape.

Guerilla is a software company building the Index of Operations, one model of a machine's parts and the connections between them that is used to monitor it, predict it and control it.

Sehaj Randhir Singh · New York

AI is now limited by physical machines, and the models watching them fail in the same four ways.

No transfer

A model trained on one motor or robot doesn't work on the next, so every unit needs its own data.

Brittle to maintenance

Swap or recalibrate a sensor and the features drift for reasons unrelated to health.

Blind at the low level

Machines fail first as a small change in how two parts relate, which big models treat as noise.

Watching and acting don't talk

One model monitors, a separate controller acts, and they share nothing.

The installed base runs for decades and almost none of it gets smarter after commissioning.

Software that tells a plant which part is going wrong, at which connection, and how early.

INSTALL

Runs on an industrial PC or edge gateway the plant already has and reads the telemetry the machine already streams. No new sensors and no hardware of ours.

WHY IT HOLDS UP

It reads how parts relate instead of raw sensor levels, so it keeps working after maintenance recalibrates a sensor, and one model carries across units of the same machine.

BUYER

The plant's reliability or maintenance manager, who owns unplanned downtime and already pays for condition-monitoring software.

The first test: one machine, one install, thirty days, against the alarms the plant already has.

A machine is parts that pass energy to each other. Watch the connections.

The index models each part (motor, pump, joint, thermal mass) and the points where parts connect. A healthy connection conserves what crosses it. When a bearing wears or an actuator weakens it stops conserving, and that signal is called slip.

$$\text{slip} = \langle\ |K_o \cdot u - a|\ /\ \max(\ K_o \cdot |u|\ ,\ |a|\ )\ \rangle$$

u = command in, a = measurement out

Slip is normalized by magnitude, so rescaling or offsetting a sensor cancels exactly. The invariance is built in, it isn't something an optimizer has to learn. The same signal tells a controller when and where to adapt.

It catches faults on real data, and it barely moves when the sensors are recalibrated.

0.951

REAL · ESA OPS-SAT

AUROC with slip added, up from the published baseline of 0.911 on the official protocol.

0.736

REAL · AFTER RECALIBRATION

Slip alone goes 0.749 → 0.736. Raw features fall from 0.919 to 0.487.

0.990

REAL · HEXAPOD ROBOT

Fault detection AUC across 154 joint-fault classes, against 0.890 for the baseline.

8.2×

SIMULATION · CONTROL

Lower tracking error under a 5× actuator fault, with frozen weights through a motor swap.

The two OPS-SAT numbers are different experiments and don't compare with each other. Full table on the site.

Every machine it reads makes the next one cheaper to bring on.

Most of physical AI is betting on collecting more hours of data. Guerilla bets on reuse.

Each deployment adds part models and connections to a shared index, so the next deployment on a similar machine is configuration instead of a new model.

That's what keeps it from being consulting, and it's why the index gets more valuable than any single install.

Retrofit monitoring for installed fleets	wedge
Controllers that survive hardware swaps	next
The grid under AI load, as certified flexible load	next
Robotics, the low-level model under robot policies	long term

The limits, said plainly, with the fix for each.

Control is simulation only. Fix: one frozen controller across a physical actuator swap on a bench motor.	next 30 days
No stability certificate on slip-gated control yet. Fix: a Lyapunov bound with the verification tools I already use.	next 60 days
Part models are hand-built, mostly rotating machinery. Fix: learn them from data.	next 90 days
No customer and no pilot yet. Fix: one design partner, one machine, thirty days.	the ask
Ten-minute SCADA averages defeat it. It needs dynamics-rate telemetry.	scope

Nine limits are listed on the site. These are the ones that matter most.

Ninety days from software on datasets to software on a real machine.

DAYS 0 TO 30

Hardware

Bench rig with two motors and a robot joint. One frozen controller across a physical swap.

DAYS 30 TO 60

Proof

Stability certificate for slip-gated control, merged into the core.

DAYS 60 TO 90

First partner

A pump and a motor in one graph, and a design partner running retrofit monitoring.

MONTHS 4 TO 12

The index grows

Edge runtime, learned part models, pilots on grid and data-center power assets.

Sehaj Randhir Singh

Junior at Queens High School for the Sciences in New York. My dad has been a grid operating engineer for 35 years, so the first machine I learned to think about was the grid, and how much of it runs on controllers tuned once and never touched again.

First author of a paper on stability proofs for learned grid controllers, accepted at the Tackling Climate Change with ML workshop at NeurIPS 2026. I built the Index of Operations solo, and five of its repositories are public.

WHAT I'M LOOKING FOR

One real machine for thirty days.

A design partner with installed equipment and telemetry, a bench to test control on hardware, and people who run machines and will tell me where this breaks.

sehajrsinghs@gmail.com
sehajr-singhs.github.io/guerilla
github.com/sehajr-singhs